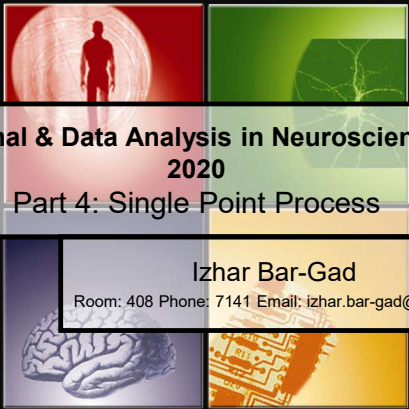



**Signal & Data Analysis in Neuroscience
2020**

Part 4: Single Point Process

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■ **Single ISI measures**

■ **Multiple ISI measures**

■ **Appendix: Real data is complicated**

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TIH/ISI

■ **ISI = Inter spike interval**

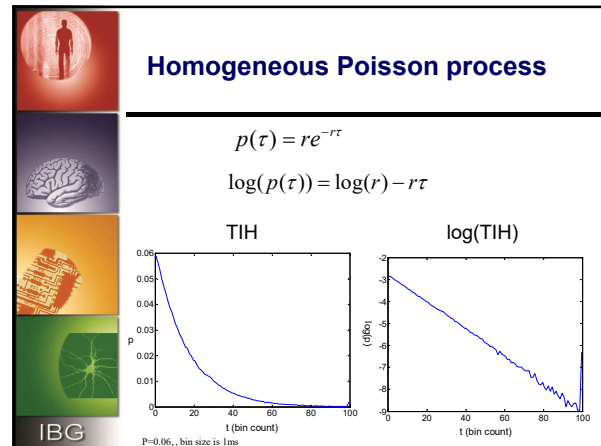
- Time difference between adjacent spikes

■ **TIH = Time interval histogram**

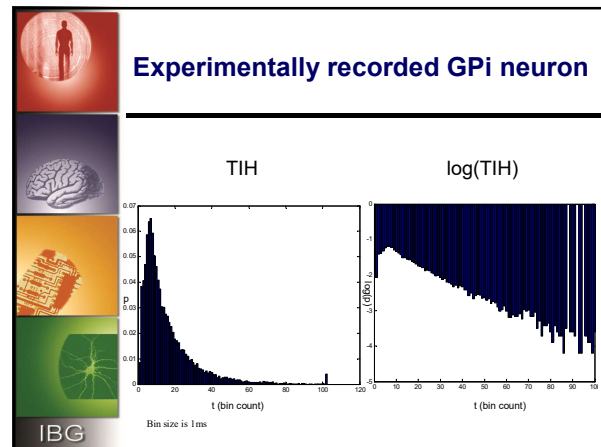
- The histogram of the time difference between adjacent spikes.

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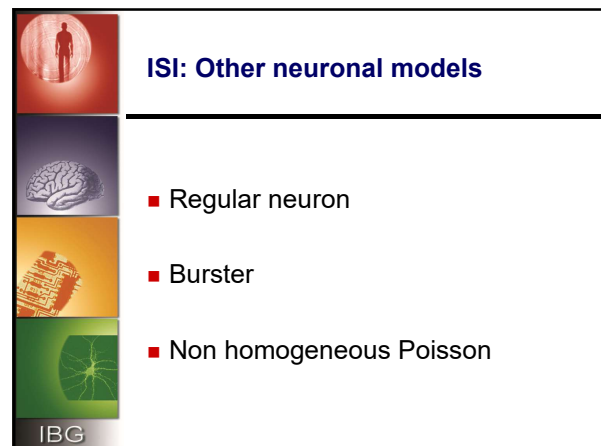
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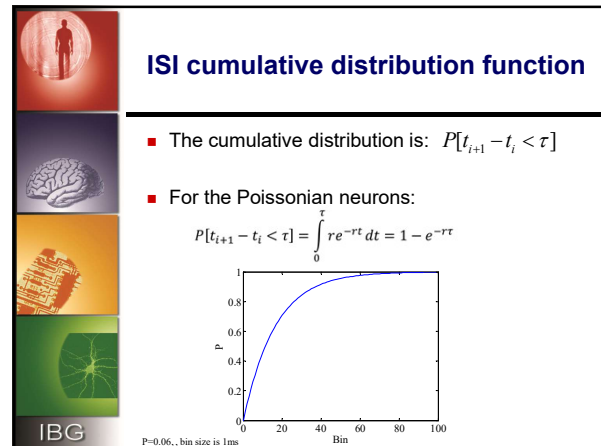
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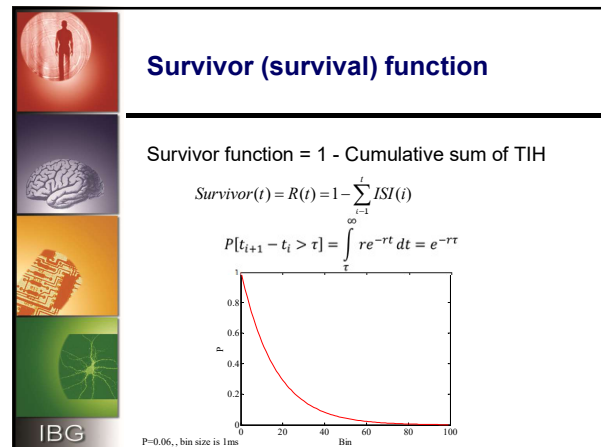
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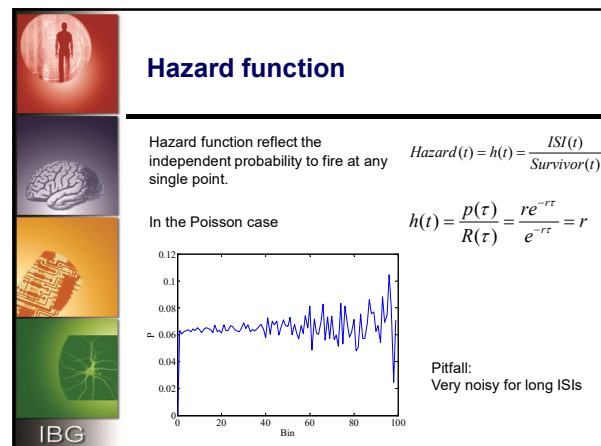
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CV – refractory period

$$CV_{noref} = \frac{\sigma_t}{\langle t \rangle}$$

- An absolute refractory period increases the mean by t_{ref} but the standard deviation remains the same

$$CV_{ref} = \frac{\sigma_t}{\langle t \rangle + \tau_{ref}}$$

- We record CV_{ref} but interested in CV_{noref}

$$CV_{noref} = \left(1 + \frac{\tau_{ref}}{\langle t \rangle}\right) \cdot CV_{ref}$$

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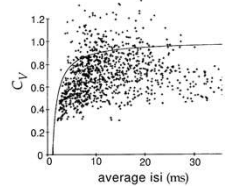
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Validation of Poisson process

Coefficient of variation



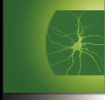
MT and V1 macaque

$$C_V = \frac{\sigma_\tau}{t_{refr} + \langle \tau \rangle} \rightarrow 1$$

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- Single ISI measures
- Multiple ISI measures
- Appendix: Real data is complicated

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Multiple ISIs

- In the Poissonian neuron, all the neuronal properties may be derived by the 1st order ISI.
- In other cases a measure of the 1st order ISI may be very different from a multi-ISI measure.
- Example: CV vs. FF of a neuron firing doublets...

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First order ISIs vs. Multi-ISI

- Shuffling → Permutation of the intervals.
- Compute the Fano factor before and after shuffling.
- If $F = F_{\text{shuffle}}$ all the irregularity may be explained by the ISIs.
- C_v remains the same...

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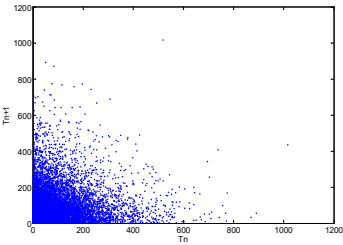
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Consecutive intervals

If consecutive interval are independent the density resembles
An exponential multiplication.

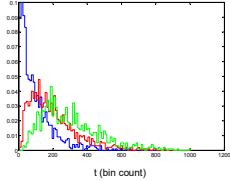


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Multiple intervals



1st (blue)
2nd (red)
3rd (green)
intervals

If x_i is distributed $\text{Gamma}(k_i, \theta)$ and x_i are independent:

$$\sum_{i=1}^N x_i = \text{Gamma}\left(\sum_{i=1}^N k_i, \theta\right)$$

Intervals are distributed $\text{Gamma}(1, \theta)$ and are independent $\rightarrow$
The n^{th} interval is distributed $\text{Gamma}(k, \theta)$

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Correlation

- Covariance

$$\text{Cov}(X, Y) = E[(X - \mu_x) \cdot (Y - \mu_y)]$$
- Pearson correlation coefficient

$$R(X, Y) = \frac{E[(X - \mu_x) \cdot (Y - \mu_y)]}{\sigma_x \cdot \sigma_y}$$

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Autocorrelation function

- General definition

$$R(t_1, t_2) = \frac{E[(X_{t_1} - \mu_{x_1}) \cdot (X_{t_2} - \mu_{x_2})]}{\sigma_{x_1} \cdot \sigma_{x_2}}$$
- Wide-sense stationary (WSS) process

$$R(\tau) = \frac{E[(X_t - \mu)(X_{t+\tau} - \mu)]}{\sigma^2}$$
- A common definition for spike trains (Dayan & Abbott)

$$Q_\rho(\tau) = \frac{1}{T} \int_0^T (\rho(t) - r) \cdot (\rho(t + \tau) - r) dt$$

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Autocorrelation practicalities

- Count $Q_C(\tau) = \sum_{i=1}^n \rho(t_i) \cdot \rho(t_i + \tau)$
- Probability $Q_P(\tau) = \frac{1}{n} \cdot \sum_{i=1}^n \rho(t_i) \cdot \rho(t_i + \tau)$
- Rate $Q_R(\tau) = \frac{1}{n \cdot \Delta t} \cdot \sum_{i=1}^n \rho(t_i) \cdot \rho(t_i + \tau)$
- Value at t=0
- Symmetry



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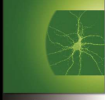

Autocorrelation practicalities

- Rate normalized version
- Count $Q_C(\tau) = \sum_{i=1}^n [\rho(t_i) - r] \cdot [\rho(t_i + \tau) - r]$
- Probability $Q_P(\tau) = \frac{1}{n} \cdot \sum_{i=1}^n [\rho(t_i) - r] \cdot [\rho(t_i + \tau) - r]$
- Rate $Q_R(\tau) = \frac{1}{n \cdot \Delta t} \cdot \sum_{i=1}^n [\rho(t_i) - r] \cdot [\rho(t_i + \tau) - r]$
- Not as common...



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Relating to standard correlation

- Covariance $C(\tau) = \frac{1}{n} \cdot \sum_{i=1}^n [\rho(t_i) - r] \cdot [\rho(t_i + \tau) - r]$
- Pearson $R(\tau) = \frac{1}{n \cdot (r - r^2)} \cdot \sum_{i=1}^n [\rho(t_i) - r] \cdot [\rho(t_i + \tau) - r]$



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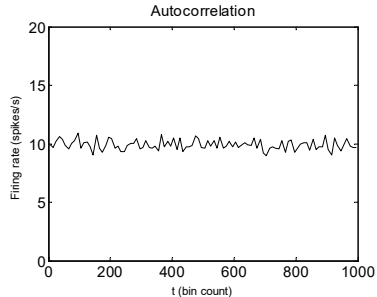
Autocorrelation practicalities

- Normalization to rate / probability / count
- Normalization to 0 vs. absolute value
- Calculating the autocorrelation:
 - All spikes at distance τ from each spike.
 - Summation of ISI of all orders.

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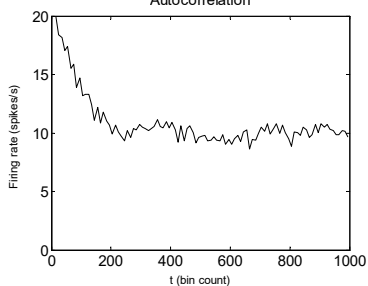
Example - Poisson



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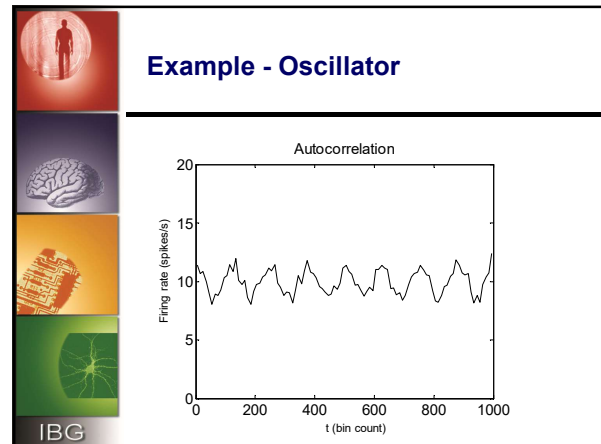


Example - Burster

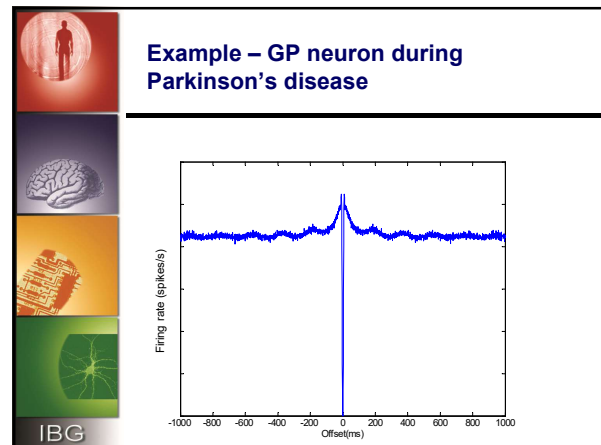


- What about a pauser?

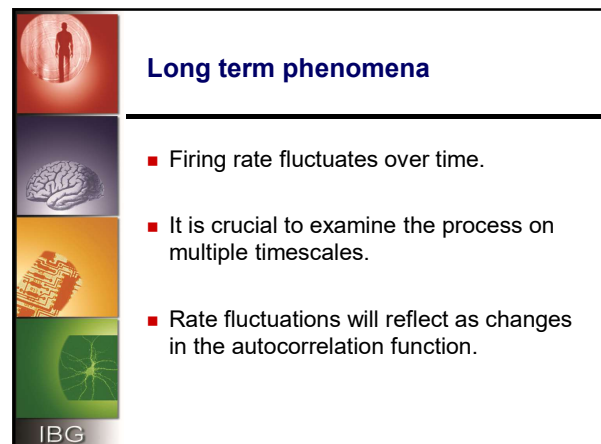
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Single spike train measures

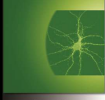
- The ISI is typically a good measure on the regularity of firing and its fit to the Poisson distribution.
- The hazard function is a good measure of short term phenomena but cannot be used on long timescales.
- The autocorrelation function is a good measure for identifying long-term phenomena.

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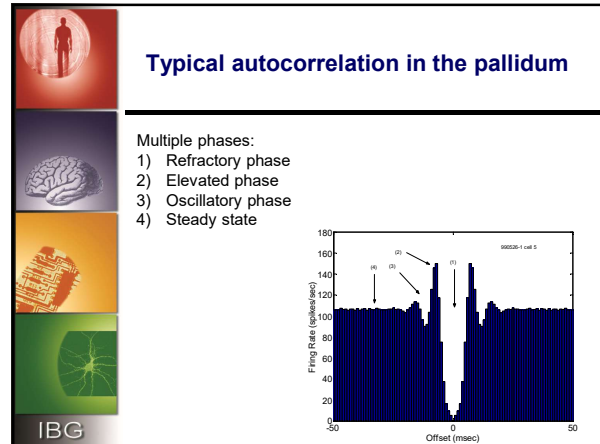
- Single ISI measures
- Multiple ISI measures
- Appendix: Real data is complicated

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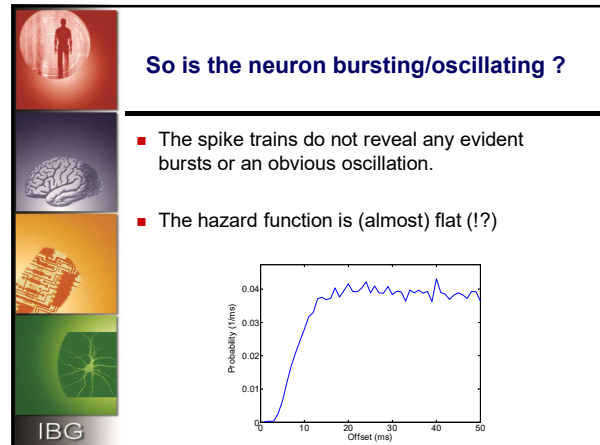

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The results of different measures are not as simple as they seem...

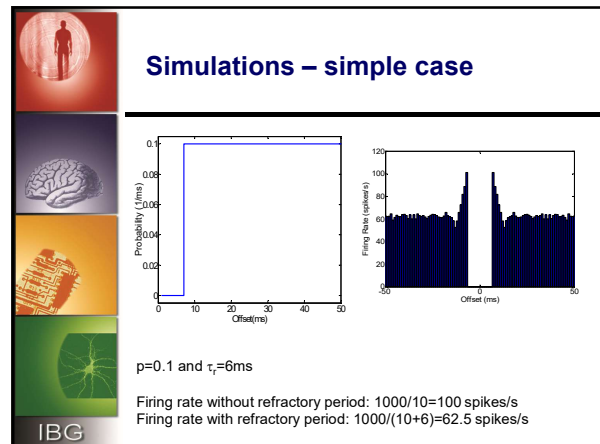
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Intuition ☺

- The key is observing the probability of being in a refractory period (RP).
- Assuming RP of length τ_r . If at any time during the last τ_r ms there was a RP then the probability for a new RP is reduced since there couldn't have been a spike during the RP.
- The autocorrelation which reflects the firing rate behavior as a negative reflection of the RP probability.



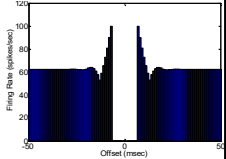
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Analysis – simple case

- (1)
$$h_t = \begin{cases} 0 & t \leq \tau_r \\ p & t > \tau_r \end{cases}$$
- (2)
$$a_t = \begin{cases} 0 & t \leq \tau_r \\ (1 - \sum_{i=1}^{\tau_r} a_{t-i}) \cdot p & t > \tau_r \end{cases}$$
- (3)
$$a_{\tau_r+1} = p$$
- (4)
$$a_\infty = (1 - \tau_r \cdot a_\infty) \cdot p \quad t \rightarrow \infty \quad \Rightarrow \quad a_\infty = \frac{p}{1 + p \cdot \tau_r}$$





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